

CSU44031 / CS7NS3 / EEU44C04 / EEP55C27

Next Generation Networks

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Course agenda

- **Three hours per week**, including lectures, tutorials, tests, invited lectures
- **Two lecturers:**
 - Marco Ruffini on the fixed network part
 - Nicola Marchetti on the wireless network part
- The **module marks** are as follows:
 - 70 % for final exam
 - 30% for course work
 - 2x tests worth 15% each

Timetable

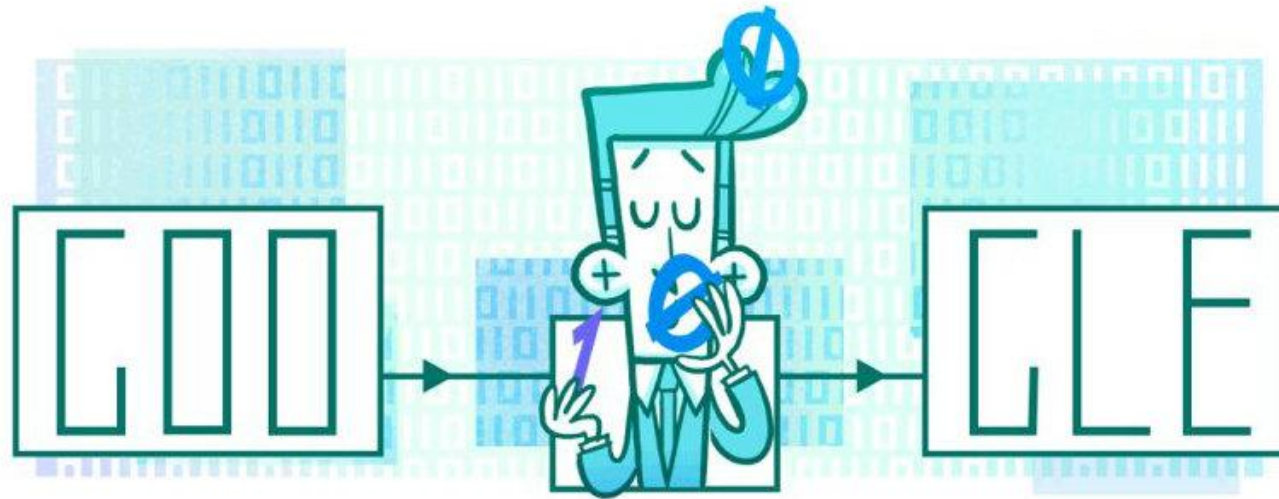
Monday, 14:00	Lloyd, LB04
Tuesday, 11:00	Lloyd, LB01
Thursday, 12:00	Lloyd, LB01

- ✓ We expect students to engage with the module from day 1 and **attend all lectures in person as per the timetable.**

Course syllabus

- Optical Transmission systems
- Contemporary and emerging wireless network architecture and systems
- Convergence of mobile and fixed architectures: backhaul, fronthaul, midhaul and protocol convergence
- Quantum communications networks and protocols
- Traffic modelling
- Recent trends in wireless networking

Claude Shannon



- https://en.wikipedia.org/wiki/Claude_Shannon
- 1914-2001
- Founder of information theory
- Founder of digital circuit design theory
- Contributed to the field of cryptanalysis (applied to codebreaking and secure telecommunications)

Hedy Lamarr

- https://en.wikipedia.org/wiki/Hedy_Lamarr
- 1914-2000
- Developed a radio guidance system, which used spread spectrum and frequency hopping technology to counter jamming
- The principles of her work are now incorporated into modern Wi-Fi, CDMA, and Bluetooth technology
- Induction into the National Inventors Hall of Fame in 2014

